

Electronic Circuit Design (EE2001)

Final Exam

Date: December 24th 2024

Course Instructor

Dr. Omer Saleem

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 5

Roll No

Section

Student Signature

Do not write below this line

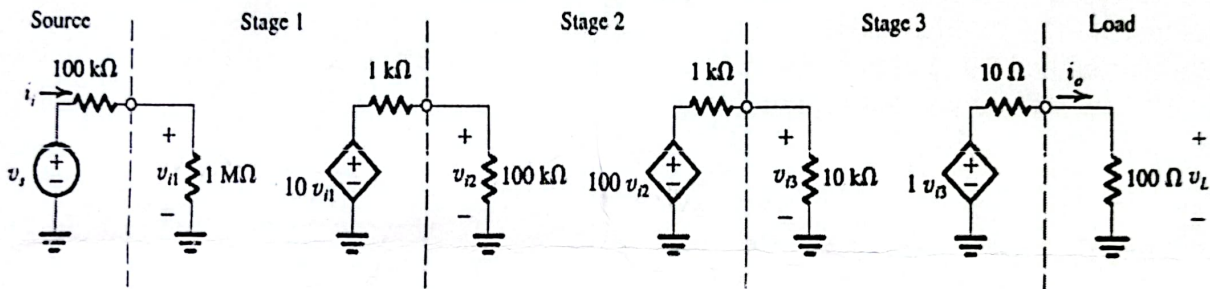
Attempt all the questions.

CLO # 1: Analyze basic amplifier circuit models.

For the circuit shown below, evaluate the overall:

[20 marks]

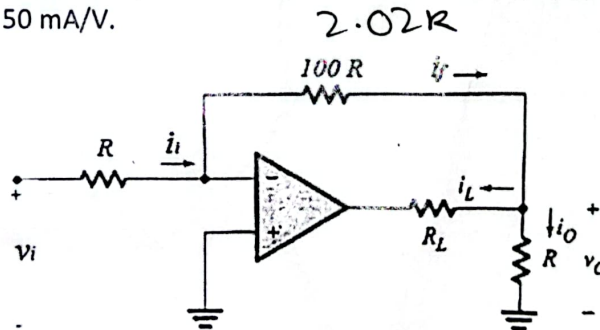
- Voltage gain, v_L/v_S , in V/V and dB. $743.727 / 57.421 \text{ dB}$
- Current gain, i_o/i_i , in A/A and dB. $8.182 \text{ M} / 138.257$
- Power gain, P_L/P_S , in W/W and dB. $6.0856 / 97.843$



CLO # 2: Design electronic circuits using op-amps.

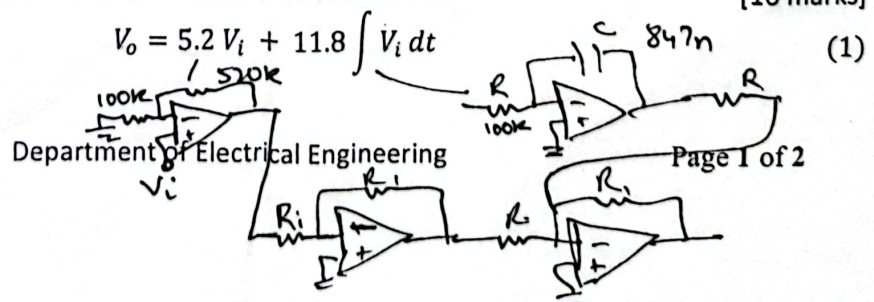
Assuming the ideal conditions, find the value of R for the following op-amp circuit whose transconductance gain, $i_L/v_i = 50 \text{ mA/V}$.

[10 marks]



Design an op-amp circuit that provides the output signal, given by Eq. 1, for an input signal V_i . Use $100.0 \text{ k}\Omega$ resistor in the input path of each op-amp circuit. Find the values of all the components used in the circuit. Clearly label and construct the circuit schematic.

[10 marks]



CLO # 3: Appraise the limitations of real op-amp circuits.

Q3 (a) A non-inverting amplifier with a closed-loop gain of 1000 V/V is designed using an op-amp having an input offset voltage of 5 mV and output saturation levels of ± 12 V. [10 marks]

Q. What is the maximum amplitude of the sine wave that can be applied to the input without the output clipping? *7 mV*

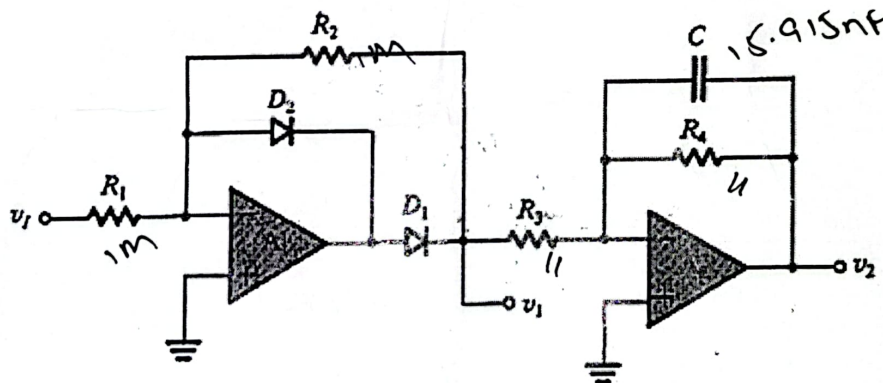
Q. What prevention must be taken against the effects of offset voltage? Draw the circuit. *-V_{os} Capacitor*

Q3 (b): An op amp circuit has a maximum output voltage ($V_{o,max}$) of 12.0 V and a slew-rate of 1.8×10^5 V/s. What is the maximum signal frequency (f_m) that can be safely handled by this op-amp? [10 marks] *2.317 kHz*

CLO # 4: Design wave-form generation circuits.

Q4 (a): Design the values of R_A and R_B for an astable multivibrator using a 555 timer IC that is capable of generating a rectangular waveform of 4000 Hz frequency and 80% duty-cycle. Use a capacitor of 0.1 μ F in your design. *R_A: 2.163 k Ω R_B: 0.71 k Ω* [10 marks]

Q4 (b): Design an AC voltmeter circuit similar to the one in the figure below which is intended to function at frequency of 10.0 Hz. It should be calibrated for sine wave input signals to provide an output of +10.0 V DC for an input of 1.0 V_{rms}. The input resistance should be as high as possible. Keep the gain in the AC part of the circuit to unity. The largest value of resistor available is 1.0 M Ω . Keep the value of capacitor as small as possible. [10 marks]



CLO # 5: Design CMOS logic circuits.

Q5 Clearly draw the CMOS implementation of $Y = AB(C + D)$ using:

- NAND and NOR gates (draw the combinational logic circuit diagram as well).
- Post-inversion technique.

